WEEK 5 REPORT

Data Visualization and Communication

1. Objective

The objective was to transform the Wine Recognition dataset into a sequence of clear visualizations and communicate the insights through a visual narrative.

2. Dataset and tools

The public Wine Recognition dataset was loaded using scikit-learn. It contains 178 samples, 13 numerical chemical measurements and three wine classes. Python, Pandas, Matplotlib and Seaborn were used.

3. Storyboard and visual methods

Chart 1 — Class distribution bar chart: compares sample counts and shows the representation of each category.

Chart 2 — Alcohol distribution histogram by class: compares the shape, center and overlap of alcohol measurements.

Chart 3 — Correlation heatmap: summarizes pairwise linear relationships between numerical chemical features.

Chart 4 — Color-intensity boxplot by class: compares medians, spread and potential outliers.

Chart 5 — Strongest pairwise correlations: a ranked table supports the heatmap by identifying the strongest positive or negative relationships.

4. Key findings

The classes contain different numbers of samples, so class-aware splitting is appropriate. Alcohol distributions can be compared visually across classes; overlap indicates that alcohol alone should not be treated as a perfect classifier. The correlation heatmap highlights groups of chemical measurements that move together. Boxplots show differences in the distribution of color intensity across classes. These visual patterns are exploratory and should be tested statistically before strong inferential claims are made.

5. Why these visualizations

Bar charts are effective for category counts; histograms reveal distribution shape; heatmaps compactly communicate many correlations; boxplots compare distributions robustly and reveal possible outliers. Clear titles, axis labels, legends and consistent class labels make the charts easier to interpret.

6. Limitations

The dataset is a small curated benchmark. Visual associations do not establish causation. Histograms depend on bin choices, correlation measures linear association, and boxplots summarize rather than show every observation.

7. Conclusion

The visual story moves from dataset composition to feature distributions, relationships and class comparisons. The charts provide an interpretable overview of the Wine Recognition dataset and support subsequent statistical modeling. The accompanying notebook recreates each visualization and includes interpretation prompts.

8. References

UCI Machine Learning Repository — Wine dataset (https://archive.ics.uci.edu/dataset/109/wine).

Matplotlib and Seaborn documentation.